

Mohamad Ammar Mohamad Hassan

Data Science Undergraduate | AI & Full-Stack Solutions Builder

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Professional Profile

Data Science undergraduate with practical experience building AI, machine learning, mobile, and full-stack applications. Skilled in Python, SQL, model development, data analysis, cloud deployment, and modern web technologies. Recent work includes a deployed retrieval-augmented generation (RAG) application, a native Android clinical calculator, and a governed multi-agent AI system. Combines strong technical execution with leadership and mentorship experience supporting 60+ student teams.

Technical Skills

Programming	Python, SQL, C, Kotlin, TypeScript, JavaScript
Data & ML	Machine learning, XGBoost, scikit-learn, feature engineering, data analysis, data visualization, SHAP
AI Systems	RAG, hybrid retrieval, LLM applications, multi-agent systems, voice AI pipelines, human-in-the-loop design
Web & Backend	React, Vite, Next.js, Tailwind CSS, FastAPI, Django REST, Node.js, tRPC, MERN
Mobile	Jetpack Compose, MVVM, CameraX, ML Kit OCR
Data & Cloud	PostgreSQL, MySQL, ChromaDB, Google Cloud Run, Docker
Professional	Problem solving, technical communication, leadership, mentorship, collaboration

Education

Bachelor of Data Science <i>Universiti Antarabangsa Albukhary (AIU), Malaysia</i>	Oct 2024 – Oct 2027
Current CGPA: 3.76 / 4.00. Relevant coursework: Data Processing, Machine Learning, Database Systems, Full-Stack Development. Activities: Treasurer, Sri Lankan Student Community; CNL Leadership Development Programme; University Sports.	
Foundation in Computing <i>Universiti Antarabangsa Albukhary (AIU), Malaysia</i>	Oct 2023 – Oct 2024

Current Projects

AI-Powered Student Success & Early Intervention Platform <i>Final Year Project XGBoost, SHAP, FastAPI, React, PostgreSQL</i>	2026 Ongoing
- Designing a weekly-cycle pipeline spanning data ingestion, risk modelling, explainability, decision rules, and advisor-facing application workflows. - Pairing an XGBoost risk model with SHAP explanations so recommendations remain auditable before advisor review. - Implementing a rule-based decision layer for safe, human-reviewed escalation rather than fully automated intervention. - Architecting role-based access for students, advisors, and administrators with a FastAPI backend and PostgreSQL data layer on Google Cloud Run.	
Kafe Eman — Multi-Branch Café Web Platform <i>React, Vite, TypeScript, Tailwind CSS, Framer Motion, Convex, Clerk</i>	2026 Ongoing
- Building a multi-branch platform covering public menus and reservations, customer checkout and loyalty workflows, and an administrative dashboard. - Implementing separate customer and staff authentication flows using Clerk and a local admin login. - Designing an admin-reviewed feedback pipeline for publishing customer testimonials to the homepage. - Planning an AI assistant layer with Groq orchestration and Twilio/Vapi-based order notifications.	
LocalVoice CRM — Privacy-First Voice-Driven CRM <i>Node.js, React, tRPC, MySQL, Ollama, Whisper.cpp</i>	2026 Ongoing
- Building a self-hosted CRM that converts spoken commands into structured actions through a local Whisper speech-to-text and Ollama LLM pipeline. - Enforcing human approval before any contact, note, or follow-up is written to the database. - Supporting up to five chained actions per utterance with a reviewable transcript and editable proposal step. - Adding optional HubSpot synchronisation while preserving offline operation with zero cloud dependency.	

Selected Projects

- AI Employee — Inbound Revenue Command Center

Autopilot Asia Hackathon 2026 | Supervity Auto, Python, FastAPI, PostgreSQL, Next.js, Docker

- Developed a governed and observable AI revenue-capture system for the Sales Intelligence track.
 - Designed an orchestrator coordinating five or more specialised AI operators across revenue workflows.
 - Implemented policy checks, human review, and exception handling for lead matching, intent-based outreach, consent conflicts, and territory allocation.
 - Built a live command centre and integrated the system through a FastAPI backend.
- DocMind AI — RAG-Based Document Q&A System

React, Vite, FastAPI, ChromaDB, Groq, Docker, Google Cloud Run

- Built and deployed a full-stack application for natural-language conversations with uploaded PDFs, including page-level citations, confidence scoring, and multi-document querying.
 - Developed a hybrid retrieval pipeline combining BM25 keyword search (40%) and ChromaDB vector similarity (60%), followed by LLM-based reranking.
 - Implemented server-sent event response streaming, evaluated retrieval quality with RAGAS, and added voice input and text-to-speech functionality.
- TDM Insight — Vancomycin Therapeutic Drug Monitoring Calculator

Kotlin, Jetpack Compose, CameraX, ML Kit OCR, MVVM

- Developed a native Android app for hospital pharmacy calculations across three Vancomycin workflows: pre-dose, post-dose, and pre/post-dose.
 - Built a standalone pharmacokinetic engine with step-by-step results for Ke, half-life, Vd, clearance, trough, and AUC24, supported by 30+ validation rules.
 - Added on-device OCR for laboratory reports with mandatory user review, local history for 20 records, and two-page PDF export and sharing.
- AI vs. Cricket — 2026 T20 World Cup Predictor

Python, XGBoost, Streamlit

- Developed an end-to-end XGBoost system that generates match-win probabilities using team performance, recent form, and venue-specific features.
 - Simulated the tournament from group stage to final and presented predictions through an interactive Streamlit application.

Leadership & Experience

- Software Lead — AquaSense, Hult Prize National Finalist

2025 – Present

- Leading development of AI-driven colorimetric analysis and LoRaMesh communication protocols.
- Student Mentor — Kedah Maker Fair 2025 Young Innovators Challenge

Aug 2025

- Mentored 66 teams from 20 schools developing AI-for-social-good prototypes in healthcare, education, and sustainability.
 - Guided students in developing, communicating, and presenting technical solutions; completed the Train the Trainer Workshop by Chumbaka Sdn Bhd.
- Project Facilitator — Net Zero Carbon Emission for Youth Leaders Programme

Mar 2024

- Coordinated youth engagement activities and workshops focused on sustainability, carbon neutrality, and climate action.

Achievements, Training & Languages

Recognition	Hult Prize AIU 2025/26 Inter-University Competition — National-level finalist with Team AquaSense.
Certifications	Google Cloud — Cloud Computing Fundamentals; Python Programming; Machine Learning Fundamentals.
Training	Full MERN Stack Bootcamp; Django REST Bootcamp; Modern Web Development Workshop: Frontend Edition; Train the Trainer Workshop.
Leadership	Treasurer, Sri Lankan Student Community — budgeting, financial management, and event coordination.
Languages	Tamil (Native); English (Fluent).
Interests	Football, cricket, frisbee, badminton, and competitive programming.

Reference

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